

UTKARSH PATHAK

Software Engineer | Backend Systems | Distributed Systems | Infrastructure

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EDUCATION

GLA University, Mathura

2023 – 2027

B.Tech in Computer Science (AI & ML, IIOT)

CGPA: 7.4/10

TECHNICAL SKILLS

Languages	Java, TypeScript, JavaScript, Python, SQL
Backend	Node.js, Express.js, REST APIs, JWT Authentication, API Gateways
Databases	PostgreSQL, Redis, MongoDB, MySQL
Distributed Systems	Apache Kafka, Caching, Rate Limiting, Event-Driven Architectures, Asynchronous Processing
Infrastructure	Docker, Kubernetes, Linux, Docker Compose, Load Testing (k6)
Systems Programming	Lexical Analysis, Parsing, AST Construction, Runtime Systems, Interpreters
Tools	Git, GitHub, Postman

PROJECTS

LinkSnip — Distributed URL Shortener

React, Node.js, PostgreSQL, Redis, Kafka, Docker, Kubernetes, k6

- Built a distributed URL shortener with Base-62 short-code generation supporting 3.5T+ unique URLs using collision-safe retry logic.
- Designed a cache-aside architecture using Redis as an L1 cache in front of PostgreSQL, reducing redirect latency through in-memory lookups with graceful fallback on cache misses.
- Implemented an asynchronous analytics pipeline using Apache Kafka, decoupling click-event logging from the redirect request path to keep redirects non-blocking under load.
- Containerized services with Docker Compose and deployed to Kubernetes with Horizontal Pod Autoscaling (2–8 replicas) for traffic-based scaling.
- Load-tested mixed redirect/create/analytics workloads using k6 (95 concurrent VUs), achieving 45ms p95 latency and 0% error rate across 7,500+ requests.

API Gateway

TypeScript, Node.js, Redis, Kafka, PostgreSQL, Docker, React

- Built a production-style API gateway featuring JWT authentication, API-key authorization, round-robin reverse proxying, and atomic Redis Lua-script token-bucket rate limiting.
- Designed an event-driven analytics pipeline (Kafka → batch consumer → PostgreSQL), batching writes in 50-row/2-second flush windows to minimize database overhead.
- Load-tested the gateway with k6 across multiple traffic scenarios (3,583 requests, up to 70 concurrent VUs), achieving 0% error rate and 23–41ms p95 latency.
- Implemented fail-open Redis degradation handling and local Kafka buffering during broker outages to improve system resilience.

jlox — Tree-Walk Programming Language Interpreter

Java

- Built a complete interpreter for the Lox programming language implementing lexical analysis, recursive-descent parsing, AST construction, runtime evaluation, closures, lexical scoping, and control flow.
- Developed a multi-stage language processing pipeline consisting of tokenization, parsing, AST generation, and interpretation, along with an interactive REPL for program execution.

ACHIEVEMENTS

- Solved 400+ LeetCode problems with strong focus on Data Structures & Algorithms.
- LeetCode solutions accumulated 11,000+ views.
- Strong understanding of Operating Systems, DBMS, Computer Networks, Object-Oriented Programming, and Compiler Design.
- Mentored 10+ students in Data Structures & Algorithms and backend development fundamentals.

RELEVANT COURSEWORK

Data Structures & Algorithms | Operating Systems | Database Management Systems | Computer Networks |
Object-Oriented Programming | Compiler Design